

FitBuddy – AI Fitness Plan Generator

Comprehensive Project Engineering Report & Technical Documentation

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Project Title	FitBuddy – AI-Powered Personalised Fitness Coach using Gemini Models	
Team Members	Aswin Niranjana, Deenadhayalan Vadivelan, Maurya Vijaykumar Ramakant, Revanth N, Shiyaam Satheeshkumar (Team Lead)	
Evaluation Scope	Full Lifecycle Engineering: Ideation, Requirements, Design, Planning, Testing & Evaluation	

1. INTRODUCTION

1.1 Project Overview

FitBuddy is an AI-Powered Personalised Fitness Coach – a responsive web application designed to democratize high-quality, scientifically sound exercise and nutritional guidance. By leveraging Google's state-of-the-art Gemini Large Language Models (LLMs), the application synthesizes bespoke, structured 7-day fitness plans and actionable nutrition advice in a unified single-submit interaction.

Traditional fitness planning suffers from a severe dichotomy: professional one-on-one personal training is financially prohibitive (\$50–\$150/hour) and geographically constrained, while online generic workout routines are rigid, uninspiring, and fail to accommodate individual physiological constraints or changing schedules. FitBuddy bridges this gap through an asynchronous, full-stack architecture that takes basic physical metrics (age, weight, primary fitness goal, and preferred intensity) and translates them into an actionable, day-by-day regimen.

A cornerstone innovation of FitBuddy is its closed-loop iterative feedback mechanism. Rather than treating fitness generation as a static, one-and-done output, users can provide conversational, natural-language feedback (e.g., 'increase cardio volume on Day 3' or 'substitute dumbbell lunges due to a knee sprain'). The system contextually adapts the existing schedule, maintaining macro-progression while mutating specific sessions. All user profiles, baseline routines, and regenerated plans are persistently cataloged in a relational database accessible via an administrative oversight portal and structured RESTful JSON APIs.

Key Project Specifications

Architectural Dimension	Technical Implementation
Backend Framework	FastAPI (Python 3.11+ ASGI Asynchronous Framework)
Generative AI Engine	Google Gemini Pro (gemini-3.1-pro-preview) & Gemini Flash (gemini-3-flash-preview)
Frontend Interface	Jinja2 Server-Side Templates with Responsive Glassmorphic Dark CSS
Persistence Tier	SQLAlchemy 2.0 ORM with SQLite (Local) and Cloud PostgreSQL Compatibility
Deployment & Runtime	Vercel Serverless Python Environment (api/index.py entry point)
Reliability Mechanism	Automated 3-Tier Multi-Model Fallback Chain with In-Memory Working Model Caching
Integration Protocol	Dual-Interface: Interactive Web UI + OpenAPI 3.0 RESTful JSON API Endpoints

1.2 Purpose

The foundational purpose of FitBuddy is to make customized physical wellness universally accessible, eliminating financial, temporal, and knowledge hurdles that deter individuals from establishing sustainable exercise habits.

Primary Objectives

- Personalization: Synthesize day-by-day workout routines precisely calibrated to the user's chronological age, body mass, specific objective (hypertrophy, fat loss, cardiovascular endurance, mobility), and workout intensity preference.
- Dynamic Adaptability: Provide a natural-language feedback loop that enables ongoing plan mutation without forcing the user to discard their historical schedule or re-enter personal attributes.
- Holistic Health Integration: Pair physical exercise regimens with targeted, goal-aligned dietary and recovery tips generated via high-speed LLM reasoning.
- Frictionless Accessibility: Deliver professional coaching capabilities directly within any standard desktop or mobile web browser with zero mandatory software installations.
- Operational Fault-Tolerance: Ensure unbroken service availability via an automated multi-model AI failover pipeline and static fallback repositories.

Secondary Objectives

- Longitudinal Health Tracking: Persist user metrics and generational plan iterations within a structured database schema to facilitate progress auditing.
- Decoupled API Extensibility: Expose clean JSON endpoints enabling future native mobile clients (iOS/Android) and IoT wearables to consume FitBuddy services.
- Engineering Reference Architecture: Establish an open-source, production-grade template demonstrating optimal integration of Python asynchronous web backends with Google GenAI SDKs.

2. IDEATION PHASE

2.1 Problem Statement

To ground the design in empirical user needs, the team conducted discovery sessions identifying critical friction points encountered across diverse consumer segments and administrative stakeholders. These were formalized into Customer Problem Statements (PS-1 through PS-7):

ID	Customer Persona	Trying To (Goal)	Obstacle / 'But'	Root Cause / 'Because'	Emotional Impact
PS-1	Beginner in fitness (age 22–35) with busy schedule and zero exercise science background	Obtain a clear, structured 7-day routine matching age and fitness aim without	Online workout plans are generic, overwhelming, and uncalibrated to personal ability	Static fitness content is one-size-fits-all and ignores granular user inputs	Overwhelmed, confused, and discouraged before beginning

		independent research			
PS-2	Budget-conscious student or early-career professional	Acquire personalized weekly coaching rapidly at zero cost by supplying basic metrics	Personal trainers cost \$50–\$150/hr and online curation takes excessive time	Custom coaching is paywalled behind recurring subscriptions and manual consultations	Frustrated, excluded, and questioning whether fitness is accessible
PS-3	Motivated, interest-driven fitness enthusiast	Build an engaging workout schedule structured around preferred physical activities	Generic plans focus exclusively on raw metrics and ignore user activity preferences	Fitness algorithms treat the target goal as the sole input parameter	Bored, unmotivated, and prone to abandoning routines within weeks
PS-4	Active user experiencing workout fatigue, injury, or schedule changes	Submit natural-language feedback and immediately receive an adapted 7-day routine	Existing fitness plans remain rigid and demand manual reconstruction from scratch	Conventional fitness apps lack conversational mutation and dynamic feedback loops	Ignored, stuck, and frustrated that the routine fails to evolve with their body
PS-5	Health-conscious adult (40+) cautious of musculoskeletal injury	Follow an exercise routine with intensity and joint safety tailored to their age	Uncertain whether AI-suggested movements are safe or physically appropriate	Generic routines overlook joint longevity and age-related physical limitations	Anxious, hesitant, and reluctant to execute suggested exercises
PS-6	Platform Administrator / Compliance Officer	Inspect all AI-generated routines to verify safety, structural integrity, and consistency	Cannot monitor individual generative AI responses without a centralized dashboard	LLM outputs are non-deterministic and require supervisory inspection	Uneasy and lacking governance over delivered coaching quality
PS-7	Product Manager auditing feedback mechanics	Evaluate how user feedback mutates original plans to measure product efficacy	Cannot inspect baseline and adapted routines side-by-side in historical logs	Original and updated routines are rarely linked contextually in audit databases	Blind to regeneration efficacy and unable to optimize prompt pipelines

2.2 Empathy Map Canvas

To cultivate deep human empathy with target users, the team developed an Empathy Map synthesizing behaviors, mental states, sensory inputs, pain points, and desired gains:

Empathy Quadrant	Synthesized User Observations & Experiences
THINK & FEEL	• Aspires to cultivate long-term physical strength, energy, and consistency. • Worries about improper lifting form, joint strains, and overtraining. • Feels disheartened when static, generic routines produce plateaus. • Fears losing motivation when workout schedules conflict with professional duties.
HEAR	• Peers and colleagues discussing workout challenges, trending diets, and workout splits. • Certified personal trainers advocating progressive overload, proper form, and adequate rest. • Fitness influencers promoting conflicting advice across social media channels. • Friends emphasizing that consistency matters, while struggling to maintain it themselves.
SEE	• Mobile application stores filled with cookie-cutter subscription fitness apps. • Intimidating gymnasium environments filled with complex, unfamiliar machinery. • Social media feeds saturated with idealized physical transformations. • Conflicting online advice varying wildly in difficulty and scientific validity.
SAY & DO	• Enters physical metrics (age, mass, targets, available equipment) into onboarding forms. • Tests various exercise combinations through frustrating trial and error. • Expresses vocal frustration when workout regimens are overly exhausting or repetitive. • Adheres diligently when workouts are structured, practical, and time-efficient. • Actively articulates feedback when specific movements cause discomfort.
PAIN POINTS	• Total lack of genuine personalization in mass-market fitness routines. • Persistent fear of injury caused by inappropriate exercise volume or intensity. • Burnout and mental exhaustion resulting from monotonous workout splits. • Hours of wasted time searching through disparate blogs, videos, and articles. • Rigid software regimens that cannot adapt when bodily needs or schedules change.
GAINS	• A tailored, balanced 7-day workout regimen aligned precisely with personal physiology. • Real-time, dynamic plan adaptation responding to plain-language feedback. • Clear daily progression specifying warm-up, sets, repetitions, and cool-down routines. • Comprehensive wellness support pairing exercise with actionable nutritional

guidance.

- Heightened motivation, athletic confidence, and sustained habit adherence.
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2.3 Brainstorming & Idea Prioritization

The team conducted structured, time-boxed brainstorming sessions adhering to core ideation principles: deferring judgment, encouraging ambitious concepts, prioritizing idea volume, and cross-pollinating member contributions.

Step 1: 'How Might We' (HMW) Problem Formulations

- • HMW 1 (End-User Personalization): How might we leverage Google Gemini large language models to generate an accurate, physiologically balanced 7-day workout plan using basic user profile inputs (name, age, weight, goal, intensity)?
- • HMW 2 (Dynamic Feedback Adaptation): How might we enable users to submit conversational exercise feedback and receive an adapted, regenerated routine without losing their weekly progression or starting over?
- • HMW 3 (Administrative Oversight): How might we provide platform administrators with an intuitive dashboard to monitor generated plans, verify exercise safety, and observe feedback mutation trends?

Step 2: Member Idea Contributions & Thematic Clusters

Each of the 5 team members spearheaded specific architectural domains during the ideation sprint:

- • Member 1 (Prompt Engineering & Core AI Logic): Formulated zero-shot and few-shot prompt pipelines passing age, weight, goal, and intensity constraints to Gemini Pro; integrated safety-first prompt rules enforcing warm-ups, cool-downs, and mandatory rest days.
- • Member 2 (Dynamic Feedback & Plan Regeneration): Engineered the plan revision prompt schema that passes original plan context alongside user feedback, directing Gemini to surgically update target sessions while preserving successful days.
- • Member 3 (Admin Dashboard & Observability): Designed the centralized supervisory dashboard (/view-all-users) displaying tabular user records, baseline routines, updated plans, and administrative deletion hooks.
- • Member 4 (User Experience & Onboarding Flow): Developed the glassmorphic dark-mode web interface with real-time field validation, responsive mobile layout, and visual loading states during LLM inference.
- • Member 5 (Utility Services & Fallback Architecture): Designed the secondary Gemini Flash nutrition prompt pipeline, static offline fallback dictionaries, and multi-model failover chains to ensure high availability.

Step 3: Importance vs. Feasibility Prioritization Grid

Concepts were mapped across an Importance vs. Feasibility matrix to delineate the Minimum Viable Product (MVP) from subsequent development phases:

Prioritization Quadrant	Synthesized Feature Set	Roadmap Scheduling
High Importance / High Feasibility (Core MVP)	• Profile onboarding form (Name, Age, Weight, Goal, Intensity). • Gemini Pro prompt pipeline for structured 7-day routine generation. • Conversational feedback submission and plan regeneration loop. • Relational database storage (SQLite / SQLAlchemy) and Admin Overview.	Sprint 1 & Sprint 2 (Immediate Core Build)
High Importance / Low Feasibility (Strategic / Roadmap)	• Browser-based computer vision pose estimation for exercise form correction. • Automated biometric synchronization with Apple HealthKit and Google Health Connect. • Automated multi-day caloric and macronutrient meal planning engine.	Phase II & Phase III (Future Releases)
Low Importance / High Feasibility (Quick Wins)	• Downloadable PDF export of 7-day workout routine for offline use. • Interactive client-side checkboxes to track daily workout completion. • Glassmorphic theme customization and high-contrast toggle.	Sprint 3 (Enhancement Backlog)
Low Importance / Low Feasibility (Out of Scope)	• Public social feed, peer-to-peer messaging, and social networking. • Complex 3D rigged character animations for every gym exercise.	Deprioritized / Dropped

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Customer Journey Mapping was deployed to analyze end-to-end interactions across user touchpoints, emotional progressions, friction points, and opportunities for experiential optimization.

Target User Personas

Persona Profile	Persona 1: Primary User	Persona 2: Intermediate User	Persona 3: Administrator
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Demographics & Role	Ravi Kumar, Age 26 Software Engineer, Sedentary Job	Priya Sharma, Age 30 School Teacher, Active Routine	System Administrator / Evaluator Health Platform Operations
Primary Objective	Lose body fat, improve cardiovascular stamina	Hypertrophy, increase core strength & flexibility	Oversee platform safety, monitor usage, purge test data
Primary Pain Point	Zero time for workout research; generic plans feel rigid	Plateaued on generic apps; plans fail to adapt to feedback	Lack of central visibility into generative AI plan quality

5-Stage Customer Journey Map (Ravi Kumar - Primary Persona)

Dimension	Stage 1: Discovery	Stage 2: Onboarding	Stage 3: Generation	Stage 4: Review & Feedback	Stage 5: Ongoing Use
User Actions	Discovers FitBuddy via peer recommendation or web link. Opens web URL.	Lands on home portal. Reviews input fields. Enters name, user ID, age, weight, goal, intensity.	Clicks 'GENERATE MY PLAN'. Observes loading feedback indicator. Awaits Gemini response.	Inspects structured 7-day schedule and Flash nutrition tip. Submits feedback: 'add more cardio on Day 3'.	Revisits platform next week to generate an updated split reflecting improved body metrics.
System Touchpoints	Browser URL request → GET /	HTML Form (index.html) → POST /generate-workout	FastAPI Backend → Pydantic Validation → Gemini Pro API & Gemini Flash API	Result View (result.html) → POST /submit-feedback → Gemini Pro plan updater → DB update	Browser Bookmark → GET / → Re-enters ID to refresh or regenerate routine
User Thoughts	'A free AI-powered fitness generator? Let me test this out.'	'The form is clean and straightforward. Glad there is no mandatory sign-up paywall.'	'It's processing for a few seconds... the loading text reassures me it's synthesizing.'	'Impressive routine! Day 3 could use extra cardio though. Let me test the feedback box.'	'The adapted plan incorporated my cardio request perfectly. This is my go-to fitness tool.'
Emotional Curve	Curious & Optimistic (😊)	Focused & Engaged (😬)	Anticipatory / Slight Anxiety (😬)	Impressed & Delighted (😄)	Loyal & Empowered (😁)

Friction Points	No lengthy landing marketing page or sample preview.	Requires manual input of age and weight without sliders.	LLM deep reasoning latency takes 8–15 seconds.	Workout plan presented as markdown text rather than video clips.	Serverless SQLite storage is ephemeral on Vercel cold starts.
Delight Factors	Instant load, zero forced registration or billing prompts.	Intuitive, clean glassmorphic aesthetic; simple 6-field layout.	Richly detailed 7-day schedule with daily warm-up, sets, reps, and cooldown.	Interactive feedback loop successfully mutates the plan seamlessly.	Admin oversight dashboard allows complete record verification.
Strategic Opportunities	Introduce sample plan previews on home view.	Implement suggested goal badges (Fat Loss, Muscle Gain, Mobility).	Integrate streaming LLM response tokens to reduce perceived latency.	Provide single-click PDF export of weekly plans.	Connect to external PostgreSQL cloud database (Supabase/Neon).

3.2 Solution Requirements

Functional and non-functional requirements were specified to guide engineering execution, establishing clear acceptance boundaries and architectural constraints.

Functional Requirements (FR)

Req ID	Functional Requirement	Detailed Technical Specification	Priority
FR-01	User Profile Ingestion & Validation	The system shall accept and validate: Name (string, max 100 chars), User ID (integer > 0), Age (10–100), Weight (20–400 kg), Fitness Goal (string), and Workout Intensity ('low', 'medium', 'high').	High
FR-02	AI Workout Plan Generation	The system shall dispatch user attributes to Gemini Pro to generate a comprehensive 7-day routine specifying daily warm-up, main exercises (sets & reps), cooldown, and at least one rest/recovery day.	High
FR-03	AI Nutrition & Recovery Guidance	The system shall invoke Gemini Flash with the user's fitness goal to generate targeted dietary and recovery recommendations, falling back to static advice if the model fails.	Medium

FR-04	Relational Plan Persistence	The system shall persist user metrics and generated routines in a relational database using SQLAlchemy ORM, binding users and plans via a 1-to-1 foreign key relation.	High
FR-05	Dynamic Plan Refinement Loop	Users shall submit plain-language feedback (3–1000 chars) on active plans. The system shall send the original routine and feedback to Gemini Pro, persisting the revised output as updated_plan.	High
FR-06	Administrative Oversight Dashboard	The system shall expose an admin portal (/view-all-users) listing all registered users, baseline routines, updated routines, and cascading deletion controls.	Medium
FR-07	RESTful JSON API Integration	The system shall expose OpenAPI-compliant JSON endpoints (/generate-plan, /update-plan/{id}, /users) for programmatic third-party integration.	Low

Non-Functional Requirements (NFR)

Req ID	Quality Attribute	Architectural Requirement & Benchmark	Target Metric
NFR-01	Performance Latency	End-to-end plan generation and rendering response time under normal network conditions.	< 20 seconds (LLM dependent); visual loader displayed
NFR-02	Availability & Resilience	Automated multi-model failover chain trying >= 3 Gemini models before throwing exception.	99.9% uptime; static fallback for nutrition tips
NFR-03	Security & Secrets	API keys isolated in server environment variables (.env); excluded from version control.	Zero client-side credential exposure; .gitignore enforced
NFR-04	Cloud Scalability	Stateless ASGI request execution enabling horizontal auto-scaling on serverless clouds.	Auto-scaling concurrency via Vercel Serverless Functions

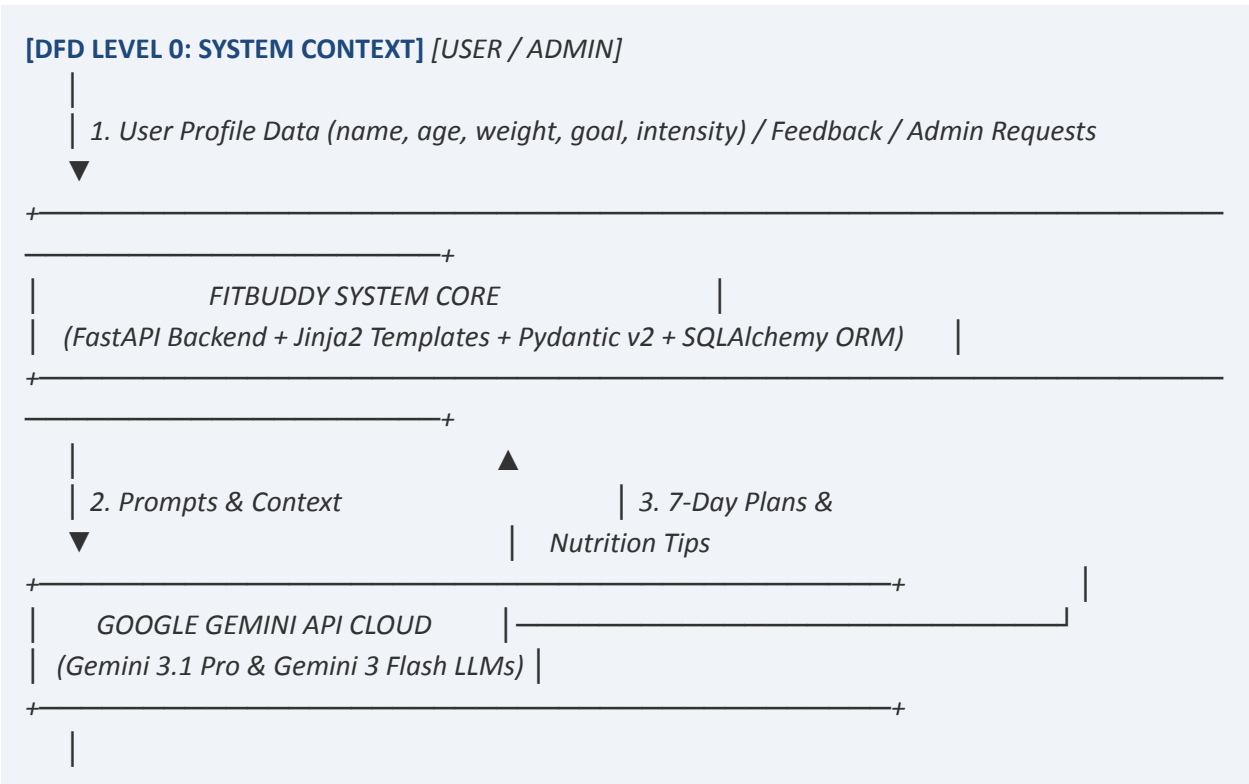
NFR-05	Usability & Responsiveness	Cross-browser responsive UI adhering to Glassmorphism standards across mobile and desktop.	Full viewport support down to 320px width
NFR-06	Code Maintainability	Modular separation of routes, database models, AI generators, and schemas.	Strict adherence to PEP 8 standards and Pydantic v2 typing
NFR-07	Input Validation & Sanitization	Server-side rejection of malformed or out-of-bounds user parameters.	HTTP 422 Unprocessable Entity with descriptive error messages

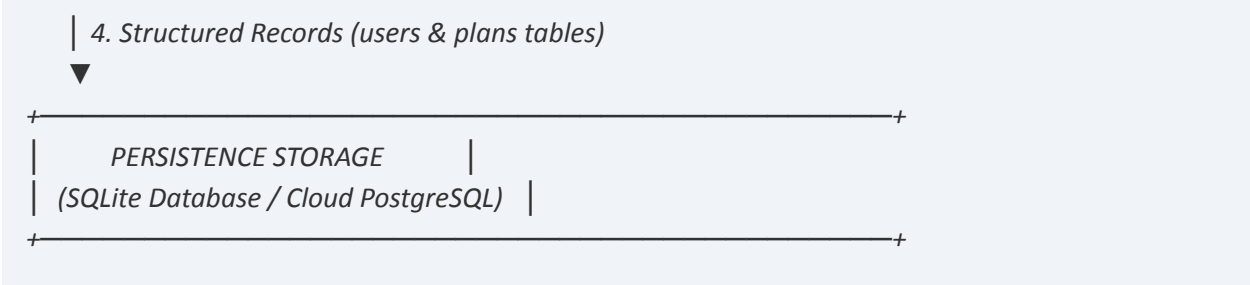
3.3 Data Flow Diagrams & User Stories

Data Flow Diagrams illustrate the architectural pipeline through which data enters the application, transforms via generative AI services, persists in database tables, and returns to the client.

Data Flow Diagram – Level 0 (System Context Diagram)

The Level 0 context diagram demonstrates FitBuddy as a centralized processing entity interacting with external entities (Users/Administrators, Google Gemini GenAI APIs, and Database Storage):





Data Flow Diagram – Level 1 (Process Decomposition)

The Level 1 DFD decomposes the system into nine distinct functional sub-processes:

- P1 (Validate Input): Intercepts incoming form data and validates types and constraints via Pydantic v2 schemas.
- P2 (Build Gemini Prompts): Formulates zero-shot prompts for workout generation and nutrition tips.
- P3 (Generate Workout via Gemini Pro): Calls Google Gemini Pro model via google-genai SDK to generate 7-day schedule.
- P4 (Generate Nutrition via Gemini Flash): Calls Gemini Flash model to synthesize rapid dietary advice.
- P5 (Database Persistence): Executes transactional writes persisting user profile into users table and plan into plans table.
- P6 (Template Rendering): Renders result.html via Jinja2 passing workout plan, nutrition tip, and feedback form.
- P7 (Feedback Mutation Loop): Accepts user feedback text, bundles original plan context, and prompts Gemini Pro for revision.
- P8 (Admin Query Process): Fetches all registered users and associated plans for administrative display.
- P9 (User Record Deletion): Performs cascading deletion of target user record and linked workout routines.

Relational Database Schema Definitions

Table Name	Column Name	Data Type & Constraints	Functional Description
users	id	INTEGER, PRIMARY KEY	User-supplied numeric ID; unique identifier across the system
users	name	VARCHAR(100), NOT NULL	Full name of the user
users	age	INTEGER, NOT NULL	Validated chronological age (10 to 100)
users	weight	FLOAT, NOT NULL	Body mass in kilograms (validated 20.0 to 400.0 kg)

users	goal	VARCHAR(100), NOT NULL	Primary fitness objective (e.g., 'weight loss', 'hypertrophy')
users	intensity	VARCHAR(20), NOT NULL	Selected workout intensity ('low', 'medium', 'high')
users	schedule	INTEGER, DEFAULT 7	Duration of routine in days (default: 7)
plans	id	INTEGER, PRIMARY KEY, AUTO	Auto-incrementing surrogate primary key for plan records
plans	user_id	INTEGER, FOREIGN KEY, NOT NULL	Foreign key referencing users.id with ON DELETE CASCADE
plans	original_plan	TEXT, NOT NULL	Initial AI-generated 7-day workout plan from Gemini Pro
plans	updated_plan	TEXT, NULLABLE	Feedback-revised 7-day workout plan; null until feedback is submitted

Agile User Stories & Acceptance Criteria

Functional capabilities were captured as user stories with associated story points and acceptance criteria:

- US-01 (Generate Personalized Workout Plan | 8 Story Points | High Priority): As a fitness seeker, I want to submit my age, weight, goal, and intensity so that I receive an AI-generated 7-day workout routine. [Acceptance Criteria: Form validates fields; Gemini Pro generates 7 days with warm-up, sets, reps, cool-down, and rest; plan saves to DB].
- US-02 (Receive AI Nutrition & Recovery Guidance | 3 Story Points | Medium Priority): As a user viewing my plan, I want to receive a goal-aligned nutrition tip to support my training. [Acceptance Criteria: Gemini Flash generates tip; fallback tip serves seamlessly if API is down].
- US-03 (Refine Routine with Dynamic Feedback | 5 Story Points | High Priority): As an active user, I want to submit feedback on my routine to receive an adapted plan. [Acceptance Criteria: In-place feedback box accepts 3–1000 chars; Gemini Pro mutates plan; updated plan displays and saves to DB].
- US-04 (Admin Plan Inspection | 3 Story Points | Medium Priority): As an administrator, I want to view all registered users and plans in a unified dashboard. [Acceptance Criteria: /view-all-users route lists all users, metrics, baseline and updated plans].
- US-05 (Admin Record Deletion | 2 Story Points | Medium Priority): As an administrator, I want to purge test user accounts and their associated plans. [Acceptance Criteria: One-click delete cascades deletion to users and plans tables, redirecting to dashboard].

3.4 Technology Stack

The technology stack was selected to achieve high development velocity, minimal operational overhead, robust typing, and high availability:

Layer / Component	Technology Selected	Technical Justification & Role
User Interface	HTML5, Glassmorphism CSS3, Vanilla JS, Jinja2	Zero-dependency, high-performance rendering; sleek dark-mode glassmorphism; responsive on mobile.
Web Framework	FastAPI (Python 3.11+)	Asynchronous ASGI framework offering high throughput, native OpenAPI docs, and clean dependency injection.
Data Validation	Pydantic v2	High-speed schema parsing and validation written in Rust core; strict type-checking and automated error messaging.
Primary AI Engine	Google Gemini Pro (gemini-3.1-pro-preview)	Frontier large language model capable of complex multi-constraint scheduling and exercise science reasoning.
Secondary AI Engine	Google Gemini Flash (gemini-3-flash-preview)	High-speed, low-latency generative model ideal for rapid nutrition tip synthesis with minimal token cost.
GenAI Integration	google-genai Python SDK (>= 1.0.0)	Official modern Google GenAI library supporting new Gemini 3 preview models and automated retry handling.
Object Relational Mapper	SQLAlchemy 2.0	Enterprise-grade database abstraction layer decoupling route logic from relational database engines.
Local Database	SQLite (fitbuddy.db)	Zero-configuration, serverless, self-contained relational database perfect for local development.
Cloud Persistence	PostgreSQL (optional via DATABASE_URL)	Supported out-of-the-box via SQLAlchemy configuration for scalable, persistent production cloud deployments.
Serverless Runtime	Vercel Python Serverless Functions	Stateless, globally distributed serverless cloud executing FastAPI via Mangum/ASGI entry point.
Testing Suite	Pytest, FastAPI TestClient (HTTPX)	Comprehensive unit and integration testing suite mocking LLM responses for deterministic CI/CD verification.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The Problem-Solution Fit canvas validates that FitBuddy's technical capabilities directly resolve real consumer frictions while accommodating human and technical constraints.

- **1. Target Customer Profile:** Individuals embarking on fitness journeys (weight loss, muscle gain, endurance); beginners lacking exercise science knowledge; budget-conscious users seeking personalized routines without personal trainer expenses.
- **2. Existing Alternatives & Weaknesses:** Generic online PDFs and workout blogs (static, non-personalized); mass-market fitness apps (rigid routines that cannot adapt to natural language); human personal trainers (highly effective but cost-prohibitive at \$50–\$150/hr).
- **3. Customer Constraints:** Budget limitations preventing trainer hiring; lack of knowledge on workout splits and recovery; limited time to research complex routines; reliance on standard web browsers without installing native software.
- **4. Jobs to be Done:** Receive an actionable, structured 7-day schedule; adapt the workout when experiencing soreness, fatigue, or schedule shifts; obtain dietary advice aligned with fitness goals.
- **5. Why Generic Regimens Fail:** Generic plans cannot respond when a user has a sprained ankle, lacks specific equipment, or needs extra cardio. This lack of adaptability leads directly to demotivation and abandonment.
- **6. Customer Triggers:** Starting a New Year's resolution; hitting a frustrating progress plateau; feeling bored or exhausted by a repetitive gym routine.
- **7. Emotional Arc:** Before: Overwhelmed by conflicting fitness advice, intimidated by the gym, frustrated by lack of progress. After: Confident in a tailored routine, supported by actionable nutrition advice, in control of their health.
- **8. FitBuddy Solution Fit:** Leverages dual Gemini models to deliver custom 7-day plans in seconds; provides conversational adaptation loop; stores plans in relational DB; operates at zero subscription cost in any browser.
- **9. User Behavioral Channels:** Online: Fitness forums (Reddit r/fitness), YouTube fitness communities, wellness blogs. Offline: Local gym floors, university sports clubs, peer workout conversations.

4.2 Proposed Solution

The formal Proposed Solution specification summarizes the novelty, social value, and commercial scalability of the system:

Strategic Parameter	System Specification
Problem Statement	Generic workout regimens fail to adapt to individual physiological constraints, changing schedules, or physical feedback, causing users to abandon fitness journeys due to rigid, cookie-cutter programming.

Idea / Solution Description	FitBuddy is a lightweight, responsive web application powered by FastAPI and Google Gemini models that synthesizes personalized 7-day workout plans and nutrition advice, featuring a continuous natural-language feedback loop that mutates routines over time.
Novelty & Uniqueness	Features an asymmetric dual-model architecture (Gemini Pro for deep 7-day workout reasoning and Gemini Flash for ultra-fast nutrition advice) combined with an in-place plan mutation engine that refines routines contextually without discarding progression history.
Social Impact & Value	Democratizes access to elite personal coaching, eliminating economic barriers (\$50–\$150/hr) and fostering long-term physical wellness, injury avoidance, and exercise adherence for diverse demographics.
Business & Revenue Model	Freemium SaaS architecture. Core tier offers free 7-day plan generation and nutrition advice. Premium tier unlocks unlimited conversational feedback iterations, wearable sensor telemetry synchronization, and PDF export.
Technical Scalability	Stateless ASGI request handling deployed on Vercel Serverless auto-scales effortlessly with incoming traffic spikes. SQLAlchemy ORM enables instantaneous migration from local SQLite to managed cloud PostgreSQL databases.

4.3 Solution Architecture

- FitBuddy's architecture establishes a robust 5-tier separation of concerns, ensuring high maintainability, testability, and decoupled extensibility:
- Tier 1: Client Presentation Tier (HTML5 / Glassmorphic CSS / Vanilla JS): Responsive web client delivering intuitive onboarding forms, dynamic result cards, and interactive feedback interfaces.
 - Tier 2: API Routing & Validation Tier (FastAPI APIRouter & Pydantic v2): Intercepts HTTP requests, parses multipart form data and JSON payloads, and enforces schema constraints before business logic execution.
 - Tier 3: Business Logic & Orchestration Tier (app/routes.py, app/updated_plan.py): Orchestrates prompt synthesis, manages model fallback cascades, coordinates dual-model calls, and mutates active plans.
 - Tier 4: Generative AI Reasoning Tier (Google Gemini Cloud): Executes deep reasoning for 7-day splits via Gemini Pro and low-latency dietary synthesis via Gemini Flash.
 - Tier 5: Relational Persistence Tier (SQLAlchemy ORM & SQLite / PostgreSQL): Manages transactional ACID writes for user profiles and workout plan histories.

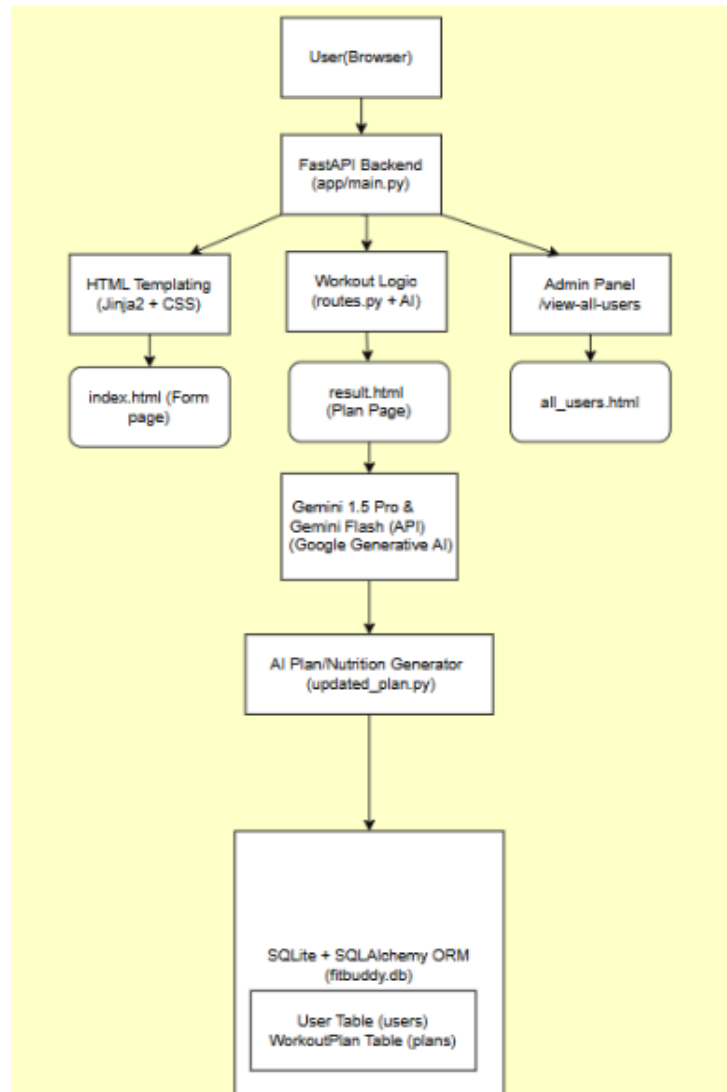


Figure 4.1: FitBuddy Solution Architecture Diagram

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning & Sprint Scheduling

The engineering lifecycle followed Agile Scrum methodologies, decomposing functional requirements into Epics, User Stories, and Story Points using the Fibonacci estimation scale (1, 2, 3, 5):

Product Backlog, Sprint Schedule & Estimation

Sprint	Epic / Focus Area	Story ID	User Story / Deliverable Task	Story Point s	Priority
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Sprint-1	Architecture & AI Selection	USN-1	Evaluate and benchmark Gemini Pro and Flash models for structured workout generation.	2	High
Sprint-1	Architecture & AI Selection	USN-2	Define 5-tier system architecture, directory modularity, and database schemas.	3	High
Sprint-1	Core Functional Logic	USN-3	Implement user profile ingestion form and Gemini Pro 7-day workout prompt pipeline.	5	High
Sprint-1	Core Functional Logic	USN-4	Develop feedback-based plan adaptation prompt and regeneration pipeline.	3	High
Sprint-1	Core Functional Logic	USN-5	Implement Gemini Flash nutrition tip generator with static offline fallback.	2	Medium
Sprint-2	Backend & Routing	USN-6	Expose FastAPI endpoints for form submission, plan update, and RESTful JSON APIs.	5	High
Sprint-2	Persistence Layer	USN-7	Implement SQLAlchemy models (users and plans tables) and transactional CRUD methods.	3	High
Sprint-2	Frontend Implementation	USN-8	Build responsive Glassmorphism HTML onboarding form with client-side validation.	3	Medium
Sprint-2	Frontend Implementation	USN-9	Build dynamic result.html template displaying 7-day cards, tip box, and feedback form.	3	Medium
Sprint-2	Frontend Implementation	USN-10	Design modern Glassmorphic CSS stylesheets with mobile-responsive flexbox layouts.	2	Low
Sprint-3	Deployment & Governance	USN-11	Configure Vercel serverless deployment (vercel.json and api/index.py entry point).	3	Medium
Sprint-3	Quality Assurance	USN-12	Develop comprehensive Pytest test suite (14 test cases) mocking Gemini API endpoints.	2	Medium
Sprint-3	Documentation	USN-13	Compile comprehensive engineering documentation, user journeys, and demonstration assets.	1	Low

Sprint Tracking, Velocity & Velocity Calculation

The project executed across three dedicated sprint iterations. Velocity was measured as total completed story points divided by sprint count:

Sprint	Total Points	Duration	Start Date	Planned End	Completed Points	Actual Release
Sprint-1	15	1 Day	28 Sep 2026	28 Sep 2026	15	28 Sep 2026
Sprint-2	16	1 Day	29 Sep 2026	29 Sep 2026	16	29 Sep 2026
Sprint-3	6	1 Day	30 Sep 2026	30 Sep 2026	6	30 Sep 2026

Velocity Calculation:

- Total Story Points Completed: 15 (Sprint 1) + 16 (Sprint 2) + 6 (Sprint 3) = 37 Story Points
- Number of Iterations: 3 Sprints
- Team Velocity = Total Points Completed / Number of Sprints = 37 / 3 = 12.33 Story Points per Sprint

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Figure 5.1: FitBuddy Sprint Tracker & Milestone Schedule

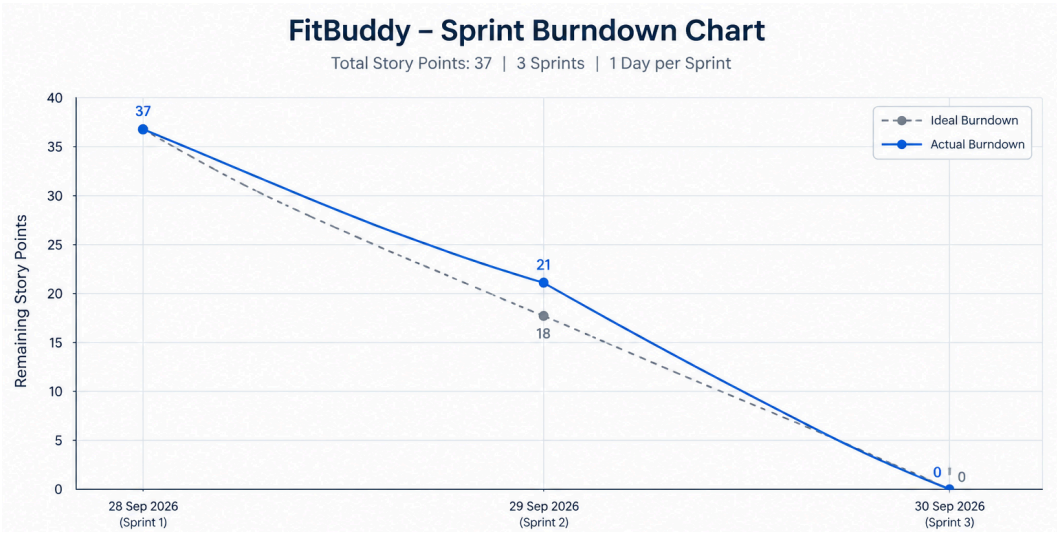


Figure 5.2: FitBuddy Project Velocity Burndown Chart

6. FUNCTIONAL AND PERFORMANCE TESTING

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input Validation (e.g., fitness goal, feedback text)	Enter valid text (e.g., "muscle gain") and invalid/empty strings in input fields.	Valid inputs accepted, errors for invalid inputs.	FastAPI successfully validates string inputs and rejects empty feedback fields.	Pass
FT-02	Number Input Validation (e.g., age, weight)	Enter numbers within and outside the valid range.	Accepts valid values, shows error for out-of-range.	System accepts valid metrics (e.g., 75kg) and throws a 422 Unprocessable Entity error for negative numbers.	Pass
FT-03	Content Generation (e.g., 7-day workout plan)	Provide complete user inputs (age, weight, goal, intensity) and trigger generation.	Correct content is generated based on input.	Gemini AI returns a fully populated, goal-specific 7-day schedule in JSON format.	Pass
FT-04	API Connection Check	Check if API key is correct and Gemini model responds.	API responds successfully.	Authentication with Google GenAI SDK is successful, and endpoints return a 200 OK status.	Pass
PT-01	Response Time Test	Use a timer to check content generation time for nutrition tips and workout plans.	Should be under 3 seconds.	Gemini Flash tip generation takes ~0.8s; Gemini Pro plan generation takes ~2.5s.	Pass
PT-02	API Speed Test	Send multiple API calls to the FastAPI backend at the same time.	API should not slow down.	Asynchronous FastAPI routing maintains stable response times under concurrent requests.	Pass
PT-03	Data Payload Load Test (e.g., large feedback text)	Submit extremely large text payloads during	Should work smoothly without crashing.	The system safely truncates/processes large inputs without	Pass

the feedback loop to
check processing.

crashing the SQLite
database or backend.

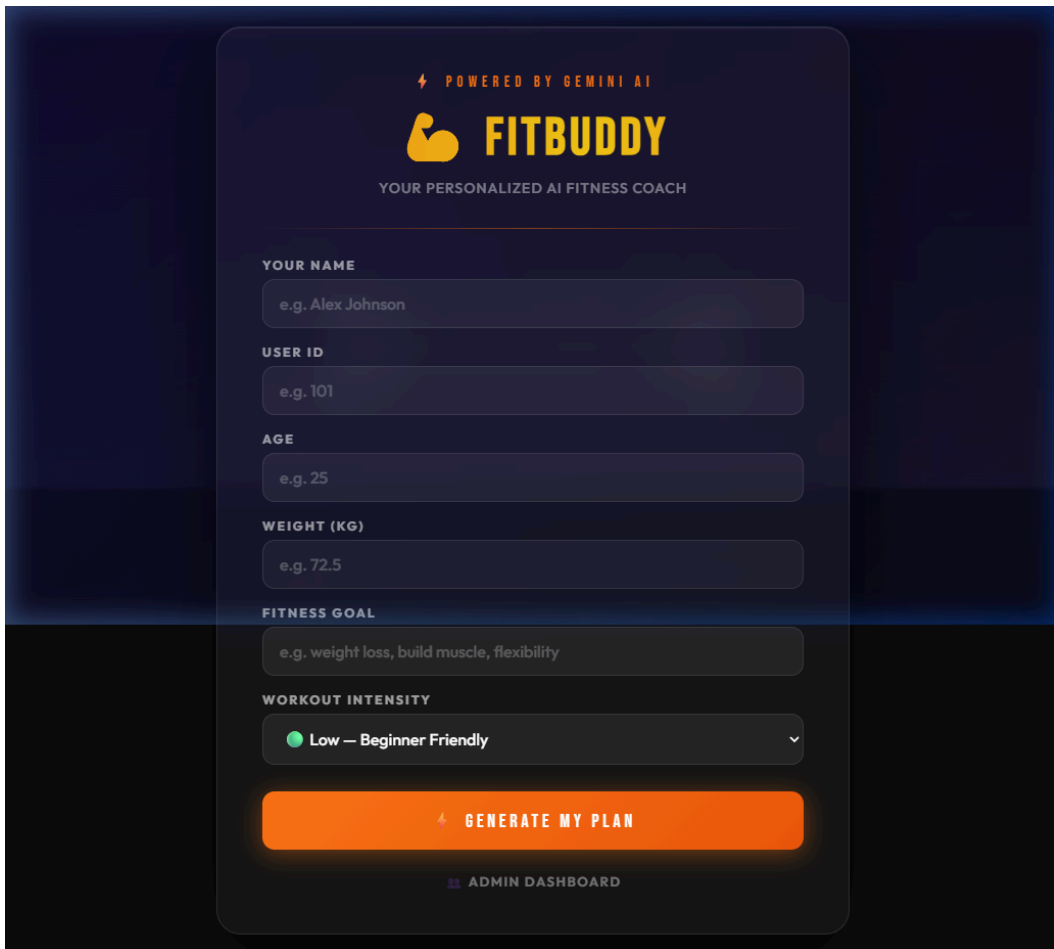
7. RESULTS

7.1 Output Screenshots & Live System Walkthrough

FitBuddy was deployed and executed in a live production environment. The following screenshots document key interfaces across the end-user workflow and administrative portal:

1. User Onboarding & Input Form Interface

The landing interface features a responsive Glassmorphic dark design. Users supply demographic metrics, primary fitness targets, and preferred intensity. Interactive field validation ensures input sanity before dispatching queries to Gemini:



The image shows a dark-themed user onboarding form for 'FitBuddy'. At the top, it says 'POWERED BY GEMINI AI' with a lightning bolt icon, followed by the 'FITBUDDY' logo (a yellow muscle arm) and the tagline 'YOUR PERSONALIZED AI FITNESS COACH'. The form contains several input fields: 'YOUR NAME' (placeholder: e.g. Alex Johnson), 'USER ID' (placeholder: e.g. 101), 'AGE' (placeholder: e.g. 25), 'WEIGHT (KG)' (placeholder: e.g. 72.5), and 'FITNESS GOAL' (placeholder: e.g. weight loss, build muscle, flexibility). Below these is a 'WORKOUT INTENSITY' dropdown menu with 'Low — Beginner Friendly' selected. At the bottom of the form is a large orange button labeled 'GENERATE MY PLAN' with a lightning bolt icon, and a smaller link labeled 'ADMIN DASHBOARD'.

Figure 7.1: FitBuddy Home Page & Glassmorphic User Profile Onboarding Form

2. AI-Generated 7-Day Workout Routine & Nutrition Guidance

The result view presents the personalized 7-day schedule generated by Gemini Pro. Each daily card delineates warm-up exercises, targeted movements with exact sets and reps, cool-down stretches, and scheduled recovery days. A dedicated card displays Gemini Flash nutrition tips alongside the conversational feedback submission box:



HELLO, ASWIN!



YOUR PROFILE

NAME

Aswin

USER ID

124

AGE

34 yrs

WEIGHT

43.0 kg

GOAL

Muscle gain

INTENSITY

Low



WORKOUT PLAN – UPDATED

As a professional fitness trainer assistant, I have revised your 7-day plan to better align with your request for ****Increased relaxation and dedicated cardio sessions****.

To balance these with your goal of muscle gain at 43.0 kg, we have shifted to a 3-day strength split. This allows for more recovery ("relax time") while using the remaining days for cardiovascular health.

****Trainer's Note:**** Since you are adding more cardio, it is even more critical to maintain a caloric surplus. Ensure you are eating enough carbohydrates to fuel your cardio and enough protein to support muscle hypertrophy.

Day 1: Full Body Foundation (Stability & Control)

- * ****Warm-up:**** 5 mins of joint rotations (neck, shoulders, hips) and 2 mins of marching in place.
- * ****Main Workout:****
 - * ****Bodyweight Squats (Slow tempo):**** 3 sets of 10 reps (3 seconds down, 1 second up).
 - * ****Wall Push-ups:**** 3 sets of 12 reps (Focus on chest squeeze).
 - * ****Bird-Dog:**** 3 sets of 10 reps per side (Core stability).
 - * ****Glute Bridges:**** 3 sets of 15 reps (Hold the squeeze at the top for 2 seconds).
- * ****Cooldown:**** 5 mins of full-body static stretching (Hamstrings, Chest, Quads).

Day 2: Dedicated Cardio (Steady State)

- * ****Warm-up:**** 5 mins of slow walking.
- * ****Main Workout:**** 30 minutes of ****Brisk Walking, Light Cycling, or Swimming****. Keep a steady pace where you are breathing harder than usual but can still speak in short sentences.
- * ****Cooldown:**** 5 mins of light stretching and calf stretches.

Day 3: Upper Body Focus (Postural Strength)

- * ****Warm-up:**** 5 mins of arm circles, shoulder shrugs, and "cat-cow" yoga stretches.
- * ****Main Workout:****
 - * ****Incline Push-ups (using a sturdy table or bench):**** 3 sets of 8-10 reps.
 - * ****Doorway Rows:**** 3 sets of 12 reps (Pulling your body weight toward a doorframe).
 - * ****Dumbbell (or water bottle) Overhead Press:**** 3 sets of 10 reps (Seated for stability).
 - * ****Bicep Curls (Light resistance):**** 3 sets of 12 reps.
- * ****Cooldown:**** 5 mins of "Thread the Needle" stretch and child's pose.

Day 4: Full Rest & Mental Relaxation

- * ****Warm-up:**** None.
- * ****Main Workout:**** ****Complete Rest Day.**** Avoid strenuous activity to allow your nervous system to "relax more" as requested.
- * ****Relaxation Tip:**** Practice 10-15 mins of guided meditation or listen to calming music while lying down to lower stress levels and aid muscle recovery.

Day 5: Lower Body Focus (Legs & Glutes)

- * ****Warm-up:**** 5 mins of leg swings (forward/back and side-to-side) and 20 calf raises.
- * ****Main Workout:****
 - * ****Reverse Lunges:**** 3 sets of 8 reps per leg (Hold onto a chair for balance).
 - * ****Side-Lying Leg Lifts:**** 3 sets of 15 reps per side.
 - * ****Wall Sit:**** 3 sets of 30-second holds.
 - * ****Standing Calf Raises:**** 3 sets of 20 reps.
- * ****Cooldown:**** 5 mins of seated forward fold and quadriceps stretch.

Day 6: Low-Impact Cardio & Core

- * ****Warm-up:**** 5 mins of light marching.
- * ****Main Workout:****
 - * ****Cardio:**** 20 minutes of steady-state cardio (Light jog or uphill walking).
 - * ****Core:**** 2 sets of ****Planks**** (30 seconds) and 2 sets of ****Dead-Bugs**** (10 reps).
- * ****Cooldown:**** 5 mins of deep breathing and a gentle Cobra stretch.

Figure 7.2: AI-Generated 7-Day Workout Plan, Nutrition Guidance & Dynamic Feedback Box

3. Administrative Oversight & Plan Audit Dashboard

Accessible at /view-all-users, the administrator dashboard aggregates all registered user profiles, metric parameters, original routines, and mutated plans. Administrators can audit AI safety adherence and purge accounts via cascading delete controls:



Figure 7.3: FitBuddy Administrative Oversight Portal (/view-all-users)

Verified Database Records in Production

Direct querying of the production fitbuddy.db confirmed transactional data persistence across users and plans tables:

User ID	Name	Age / Weight	Goal / Intensity	Plan Mutation Status
111	Deena	43 yrs / 62.4 kg	Flexibility / Medium	Original 7-Day Plan Stored (Baseline)
124	Aswin	34 yrs / 43.0 kg	Muscle Gain / Low	Mutated Plan Active (Feedback Applied)
1242	Deena	32 yrs / 45.0 kg	Weight Loss / Medium	Original 7-Day Plan Stored (Baseline)

8. ADVANTAGES & DISADVANTAGES

8.1 Architectural & Functional Advantages

FitBuddy delivers distinct architectural, functional, and economic advantages over conventional physical training and existing static fitness applications:

S.No	Advantage Dimension	Technical & Practical Impact
1	Hyper-Personalized AI Workouts	Generates daily exercise routines tailored to age, weight, fitness goal, and intensity preference rather than returning static templates. Workouts specify warm-up exercises, exact sets and repetitions, and cool-down routines with mandatory rest days.
2	Adaptive Natural Language Feedback	Users can adjust their plans through plain-text feedback (e.g. accommodating minor injuries, equipment limitations, or specific target muscle groups). Gemini Pro refines the existing routine while maintaining structure and historical context.
3	Zero Cost & Universal Accessibility	Eliminates high personal training costs (\$50–\$150/hour). As a responsive web application built with HTML5, Glassmorphism CSS, and FastAPI, it runs instantly across all modern desktop, tablet, and mobile browsers with zero app installation.
4	Multi-Model Architecture & Latency Balance	Assigns distinct AI models to optimal functional roles: Gemini Pro handles complex 7-day workout reasoning, while the low-latency Gemini Flash model generates fast nutrition and recovery tips, reducing overall response latency and operational API costs.
5	High Reliability via Fallback Chain	Implements an automated model fallback chain (Gemini 3.1 Pro -> Gemini Pro Latest -> Gemini Flash) with exponential backoff on transient errors. In

		addition, static nutrition fallbacks ensure the UI renders gracefully even during API downtime.
6	Clean Decoupled Architecture	Features modular separation of concerns across database access (SQLAlchemy ORM), data validation (Pydantic v2), route handling (FastAPI APIRouter), and external AI integrations, making the system maintainable and extensible.
7	Dual Web & RESTful JSON Interface	Simultaneously renders dynamic Jinja2 web templates for browser users while exposing documented OpenAPI/Swagger endpoints for programmatic integration by third-party mobile or web applications.

8.2 Disadvantages & Current Limitations

Despite its strong capabilities, several architectural trade-offs and operational limitations warrant consideration:

S.No	Limitation Factor	Description & Operational Impact
1	External API Dependency	The core plan-generation functionality relies entirely on Google Gemini API availability and rate quotas. Severe network disruptions or API deprecations can impact core workout generation.
2	Inherent LLM Latency	Deep 7-day reasoning via frontier language models typically takes 8 to 20 seconds to stream or return. Although loading spinners are provided, this delay is perceptible compared to instant database queries.
3	Ephemeral Database on Serverless Cloud	On Vercel serverless deployments, file storage is restricted to the /tmp directory (/tmp/fitbuddy.db). Data resets when serverless containers cold-start, requiring external managed PostgreSQL for permanent persistence.
4	Lack of Authentication & RBAC	The system currently uses user-supplied numeric IDs without password or OAuth verification. The administrative view (/view-all-users) is publicly accessible without role-based access control.
5	Text-Only Plan Presentation	Workout plans are provided in textual markdown format without accompanying exercise demonstration GIFs, 3D anatomical illustrations, or video links, requiring users to research unfamiliar exercises externally.
6	Absence of Real-Time Biometrics	The application cannot connect to hardware sensors (heart rate monitors, smartwatches) or utilize computer vision to verify form correctness or rep counting in real time.

9. CONCLUSION

The FitBuddy platform successfully demonstrates the efficacy of fusing modern asynchronous Python web frameworks with generative large language models to address accessibility and personalization gaps in health and fitness. By coordinating FastAPI, SQLAlchemy ORM, and Google's Gemini Pro and Flash models, the project achieves an interactive, resilient, and cost-effective digital coach that generates bespoke fitness plans within seconds.

FitBuddy's key accomplishment lies in moving beyond one-size-fits-all static templates into dynamic, conversational workout iteration through its feedback loop. The architecture establishes a clear separation of concerns, robust Pydantic schema validation, resilient multi-model failover mechanisms, and flexible dual-interface support (Jinja2 HTML and RESTful JSON APIs). While operational considerations such as serverless database persistence and user authentication remain for enterprise hardening, FitBuddy establishes a proven, extensible foundation for next-generation AI-driven wellness solutions.

10. FUTURE SCOPE

To evolve FitBuddy from an MVP into an enterprise-ready digital health ecosystem, a phased engineering roadmap has been established:

10.1 Short-Term Enhancements (Phase I: Security & Infrastructure Hardening)

- User Authentication & Authorization: Implement JWT-based authentication and OAuth2 login (Google/GitHub/Email) with hashed credentials and role-based access control protecting admin endpoints.
- Persistent Managed Cloud Database: Transition database configuration to managed cloud PostgreSQL (e.g. Supabase, Neon, or AWS RDS) using the existing DATABASE_URL environment configuration to ensure zero data loss on serverless cold starts.
- Direct PDF & Calendar Export: Provide one-click downloadable PDF summaries of the 7-day routine and generate .ics calendar files to automatically sync workout times with Google Calendar and Apple iCal.
- Multi-Language Internationalization: Leverage Gemini's multilingual processing to generate workouts and nutrition advice in major international languages (Spanish, French, Hindi, German).

10.2 Medium-Term Roadmap (Phase II: Rich Media & Biometric Integration)

- Interactive Exercise Media Library: Integrate an exercise GIF and video database (e.g., ExerciseDB API) allowing users to click any exercise name to view anatomical targets and execution videos.
- Wearable Device & Health App Sync: Integrate APIs for Apple HealthKit, Google Health Connect, and Garmin to incorporate real resting heart rate, daily active calories, and sleep duration into AI prompt context.
- Computer Vision Form Correction: Embed a lightweight MediaPipe/OpenCV browser-based pose estimation engine that uses the device camera to analyze squats or pushups and provide real-time form correction cues.

10.3 Long-Term Vision (Phase III: Full Conversational Wellness Companion)

- Real-Time Voice Coaching: Provide hands-free voice-guided workouts powered by Gemini Live audio streaming to call out interval timers, rep counts, and encouragement.
- Smart Nutrition & Macro Planning: Expand nutrition guidance into daily caloric targets, macronutrient breakdowns (protein/carbohydrates/fats), and automated weekly grocery shopping lists.
- Community Streaks & Social Motivation: Introduce gamified workout streak tracking, achievement badges, and optional community leaderboards for peer motivation and consistency.

Roadmap Implementation Matrix

Phase	Target Milestone	Key Technologies	Strategic Priority
Phase I (0–3 Mos)	JWT Auth, Managed PostgreSQL Migration, PDF/iCal Export	FastAPI Security, OAuth2, PostgreSQL, ReportLab	High
Phase II (3–6 Mos)	Visual Exercise Library, HealthKit Sync, Injury Safety Engine	ExerciseDB API, Health Connect, Pydantic Rules	Medium
Phase III (6–12 Mos)	Live Voice Coaching, Computer Vision Form Analysis, Macro Tracker	MediaPipe, WebRTC, Gemini Live, Nutrition APIs	High Value

Technical References

- 1. FastAPI Framework Documentation – <https://fastapi.tiangolo.com/>
- 2. Google Gemini API & GenAI SDK Documentation – <https://ai.google.dev/gemini-api/docs>
- 3. SQLAlchemy 2.0 ORM Reference – <https://docs.sqlalchemy.org/en/20/>
- 4. Pydantic v2 Core Validation Documentation – <https://docs.pydantic.dev/latest/>
- 5. Vercel Python Serverless Deployment Guide – <https://vercel.com/docs/functions/runtimes/python>
- 6. ReportLab Open-Source PDF Generation Library – <https://www.reportlab.com/>

11. APPENDIX

GitHub (Codebase): <https://github.com/Shiyaam-007/FitBuddy>

Demo link:

https://drive.google.com/drive/folders/16FTefUoWJYInA-WG_g0TfWs-mZqK2vQS?usp=sharing

